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Pico@rp2:/$ help
Available Commands:

help, h          Show this help message
wifi             Manually scan and connect to a Wi-Fi network
ifconfig         Show network interface configuration (IP address, etc.)
sysinfo          Display system info
dSPACE           Shows the available disk space on the device
ram              Shows the current RAM usage

Filesystem Commans:
pwd              Print the current working directory
ls              List files and directories
cd <dir>         Change current directory
mkdir <dir>      Create a new directory
rmdir <dir>      Remove a directory
rm <file>        Delete a file
read <file>      Prints the contents of a file

Script & Networking:
run <le.py>      Execute a Python script from storage
clone <url> [name] Download a .py or .txt file from the internet (GitHub raw supported)
curl <url>       Fetch and display content from a URL
ping <host>      Ping a host by domain or IP
scan            Scan for local networks
pmap <host>      Nmap-style port scanner that will check for open port on a designated network

Telnet:
telnet launch    Start a telnet server.
telnet stop      Stop the telnet servr.
Note: Telnet is started by default if the device is able to successfully connect to Wifi.

System:
clear           Clear the screen
about           Show system version and shell info
setclock <MHz> Set the CPU clock speed (allowed range: 40-260 MHz)

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Figure 2: PicoShell help

a Pi Pico 2 W board to play with the PyDOS as it packs many utilities that require more memory to run than available on Pi Pico 1 boards. But no worries in case you only have the Pico 1 board as you can cut the additional utilities then. Without further ado let's add a DOS-like shell running with all the functions of a 1981 IBM PC on a Pi Pico.

I found the PyDOS working out of the box better with the CircuitPython firmware on the Pico 1/2 boards. So first flash your Pico board with an appropriate .uf2 file downloaded from the official CircuitPython site. Your board should appear as a mounted flash volume on your machine after the correct flashing. Now directly copy the PyDOS files (ignore the *cpython* folder on MicroPython firmware or *mpython* folder on the CircuitPython firmware to save flash space) to the mounted Pico board after downloading its zip from the official GitHub project page and uncompressing it. If you are trying with a less memory Pico 1 board or have limited flash space, the *PyDOS.py* program will run standalone to launch the shell. So you can simply copy only *PyDOS.py* to your microcontroller to begin with. After copying the necessary file(s), just power cycle the board so that the file system is configured to allow the microcontroller to have read/write access. When you do this, the host computer will no longer be able to write to the microcontroller mounted drive.

Now connect to the board over the serial terminal as we did in the last sections. If you are not going for the full setup then the board will boot to the CircuitPython prompt. Type *import PyDOS* on the CircuitPython prompt and the DOS shell will start running with a prompt showing the available RAM amount. Now type the *dir/w* command to show the directories and files listing in wide columns mode. Entering the *mem* command shows the available RAM and performs a garbage collection operation. The *date* and *time* commands are self-explanatory. You can change the shell prompt to something more interesting using the *prompt* command. Entering the *type* command displays contents of a file.

The DOS-shell provides more commands, which enable you to create DOS-style batch file scripts. The PyDOS project page has nice documentation to cover all the commands provided. A terminal session running the DOS shell after copying only the *PyDOS.py* to a Pico 1 board is shown in Figure 3. Typing the *exit* command on the DOS-shell will drop you to the CircuitPython prompt.

Next, let's install the PyDOS in its full glory on a Pico 2 W board. Just copy all the PyDOS files (you can choose MicroPython over CircuitPython) and power cycle the board. You should boot directly into the DOS-shell when connected over the serial link. The setup command will